RAJALAKSHMI ENGINEERING COLLEGE

P Saam 251801276-2025-AIDS

P2

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|---|---|
| **Started on** | Thursday, 27 August 2026, 9:07 AM |
| **State** | Finished |
| **Completed on** | Thursday, 27 August 2026, 9:07 AM |
| **Time taken** | 34 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

**For example:**

| Input | Result |
|---|---|
| 5<br>67 34 12 98 78 | 12 34 67 78 98 |

**Answer:**

```c
#include <stdio.h>
#include <stdlib.h>

// Function to swap two elements
void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}

// Partition function using the last element as the pivot
int partition(int arr[], int low, int high) {
    int pivot = arr[high]; // Pivot element
    int i = (low - 1);     // Index of smaller element

    for (int j = low; j <= high - 1; j++) {
        // If current element is smaller than or equal to pivot
        if (arr[j] <= pivot) {
            i++; // Increment index of smaller element
            swap(&arr[i], &arr[j]);
        }
    }
    swap(&arr[i + 1], &arr[high]);
    return (i + 1);
}

// Quick Sort function (Divide and Conquer)
void quickSort(int arr[], int low, int high) {
    if (low < high) {
        // pi is partitioning index, arr[p] is now at right place
        int pi = partition(arr, low, high);

        // Separately sort elements before partition and after partition
        quickSort(arr, low, pi - 1);
        quickSort(arr, pi + 1, high);
    }
}

int main() {
    int n;
    // Read the number of elements
    if (scanf("%d", &n) != 1) return 0;

    int *arr = (int *)malloc(n * sizeof(int));

    // Read the array elements (each on a new line or space-separated)
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    // Call Quick Sort
    quickSort(arr, 0, n - 1);
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>67 34 12 98 78 | 12 34 67 78 98 | 12 34 67 78 98 | ✔ |
| ✔ | 10<br>1 56 78 90 32 56 11 10 90 114 | 1 10 11 32 56 56 78 90 90 114 | 1 10 11 32 56 56 78 90 90 114 | ✔ |
| ✔ | 12<br>9 8 7 6 5 4 3 2 1 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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